

Regression to the Mean

Fun Fridays · Professional Minds · Week 5

Did the intervention cause improvement, or did an extreme result naturally move closer to average?

Recognize stable factors plus chance, and demand comparison before causal stories.

Today's finish line

DO THIS See the core move.

WHAT THIS MEANS

When selection begins at an extreme, expect some movement toward average even without treatment.

DO THIS Try it on a bounded example.

WHAT THIS MEANS

A manager coaches the five lowest-performing reps; performance rises, but no comparison group tests whether coaching caused it.

DO THIS State evidence and limits.

WHAT THIS MEANS

Selection rule, baseline, repeated measure, comparison/control, and a causal limit.

What does this prove, and what does it not prove?

DO THIS → WHAT THIS MEANS → FIND THIS YOURSELF → DEMAND DOCUMENTATION
VERIFY ON YOUR MACHINE → KEEP THE RECEIPT

The mental model

The idea

Regression to the mean is expected when outcomes combine stable factors with chance and cases are selected for extreme results.

The boundary

The pattern is real; its popular causal story may be wrong. Regression does not prove an intervention useless.

FIND THIS YOURSELF Before accepting a conclusion, name the observation that would support it.

A worked computing example

DO THIS Apply the idea to this bounded case: A manager coaches the five lowest-performing reps; performance rises, but no comparison group tests whether coaching caused it.

First pass

- ▶ What is known now?
- ▶ Which step or assumption is visible?
- ▶ Where is the checkpoint?

Professional move

Compare selected reps with similarly low performers who did not receive the program, or state exactly what the before/after design cannot show.

Label the application: This is an instructor/computing translation, not a claim about the primary source.

Did the intervention cause improvement, or did an extreme result naturally move closer to average?

Predict

What would you expect if the first story were incomplete?

Compare

What alternative explanation deserves a fair test?

PAIR TALK: Make one claim, one reason, and one uncertainty visible.

Do this: make the move visible

DO THIS Write what selected the cases, what stayed stable, what could be chance, and what comparison would sharpen the claim.

1. Write the task, decision, or claim in one sentence.
2. Mark the relevant stages, evidence, comparison, or criteria.
3. Choose one checkpoint before you act or explain.
4. Say what would make you revise, stop, or continue.

VERIFY ON YOUR MACHINE A good first attempt leaves a receipt another person can inspect.

What the source supports — and its limits

Supports

- ▶ Stable factors plus chance, selection on extremes, and comparison/control logic explain why before/after stories mislead.
- ▶ A clearer question and a more bounded next move.

Does not prove

Regression alone does not identify every cause; historical examples need careful source and limitation handling.

Is this safe? Is this secure?

How do I know? Where can I verify that?

Recovery: when the first story fails

PAUSE

A failed check is information about the route.

Use one bounded recovery

1. Preserve the observation and context.
2. Name the assumption or step to inspect.
3. Make one smaller retry or ask for help.
4. Do not turn uncertainty into a confident story.

Watch for: Crediting a cure for natural improvement, or treating regression as a mysterious force rather than a statistical pattern.

Keep the receipt

DO THIS Record the smallest useful evidence.

- ▶ Claim or decision
- ▶ Observation and context
- ▶ Reasoning / comparison / checkpoint
- ▶ Result, limit, and next action

Exit move

Find one before/after claim and add the missing comparison, control, or causal limitation.

KEEP THE RECEIPT A concise receipt beats a confident story.

Your next move

Find one before/after claim and add the missing comparison, control, or causal limitation.

How will you know?